

R0004E Exercise 2

1.

Given the transfer function of Eq. (4.23), find ζ and ω_n .

$$G(s) = \frac{36}{s^2 + 4.2s + 36}$$

2.

PROBLEM: Given the pole plot shown in Figure 4.20, find ζ , ω_n , T_p , %OS, and T_s .

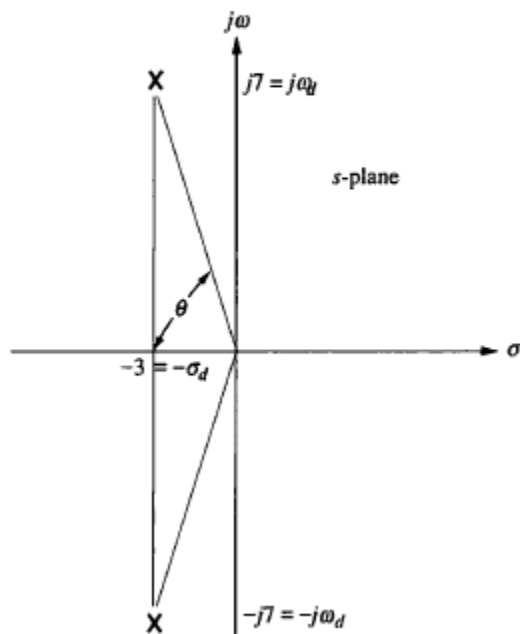


FIGURE 4.20

3.

PROBLEM: Convert the state and output equations shown in Eq. (3.78) to a transfer function.

$$\mathbf{x} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t) \quad (3.78a)$$

$$y = [1.5 \quad 0.625] \mathbf{x} \quad (3.78b)$$

Compare the stability of the system in state-space and in transfer function forms.

4. Given a plant with the transfer function

$$G(s) = \frac{36}{s^2 + 4.2s + 36}$$

To improve the stability of this system, which element(s) of PID controller needs to be applied to this system?

5. Given a plant with the transfer function

$$G(s) = \frac{36}{s^2 + 24s + 36}$$

Find the steady state error for the unity feedback system with respect to a unit step input.

To improve the response of this system, in terms of steady-state error, which element(s) of PID controller needs to be applied to this system?